ARUN AGARWAL

arun.agarwal@berkeley.edu | 484-787-8461 | https://linkedin.com/in/arunagarwal23 | https://github.com/aagarwal17 | West Chester, PA

EDUCATION

University of California, Berkeley, School of Information *August 2026*
Master of Information and Data Science (MIDS) **GPA: 4.0/4.0**
Selected Coursework: Introduction to Data Science Programming, Research Design and Applications for Data and Analysis, Statistics for Data Science, Fundamentals of Data Engineering, Applied Machine Learning, NLP with Deep Learning, ML at Scale

Temple University, College of Science and Technology, University Honors Program, Philadelphia, PA *May 2023*
Bachelor of Science in Data Science | Bachelor of Science in Math and Computer Science | Certificate in Spanish **GPA: 3.99/4.0**
Selected Coursework: Linear Algebra, Statistical Business Analytics, Program Design and Abstraction, Introduction to Python, Computer Systems and Low-Level Programming, Data Structures and Algorithms, Principles of Data Science, Probability Theory, Applied Statistics and Data Science, Systems Programming and Operating Systems, Knowledge Discovery and Data Mining, Foundations of Machine Learning, Software Design, Numerical Analysis, Database Systems, Capstone in Data Science

TECHNICAL SKILLS

- **Languages**: Python (Pandas, Matplotlib, Sklearn, PySpark, Numpy, Tensorflow, etc.); R (Tidyverse, Shiny); C; Java; LaTeX; SQL; NoSQL; HTML
- **Software/Services**: Jupyter Notebook, IntelliJ, PyCharm, Eclipse, VSCode, IDLE, RStudio, AWS (S3, Glue, Athena, CloudWatch, Lambda, SageMaker, DynamoDB, API Gateway, ECS, EC2, etc.), Tableau, Git, Jira, JMP Pro, Docker, Anaconda, Code Lite, Code::Blocks, Jupyter Lab, Excel, Postman, CoPilot, Oracle SQL Developer, Postgres, Neo4j, MongoDB, Redis, Kore AI, Apache Spark, MATLAB, Bitbucket, Power Query, Honeycomb, Splunk, Genesys, PagerDuty, Coder, etc.
- **Skills**: Data Wrangling, Data Visualization, Web Scraping, Statistics, iOS, Technical Support, Testing (Unit, Integration, Locus, Cyara), Scrum, Web APIs, Data Engineering, Natural Language Processing (NLP), Generative Adversarial Networks (GANs), Cloud Computing, CI/CD (Continuous Integration/Continuous Development), Test Driven Development, Generative AI and LLMs, API onboarding and integration, Prompt Engineering, Backend development, Quantum Computing, Data Engineering, Machine Learning, etc.
- **Certifications**: AWS Certified Cloud Practitioner – *December 2023*, Kore AI Automation Training – *August 2024*, IBM Qiskit Quantum Computing – *October 2024,* Professional Scrum Master I (PSM I) (*in progress*), AWS Machine Learning Associate (*in progress*)
- **Conferences**: Open Data Science Conference (ODSC East 2025): ODSC AI Bootcamp + 6-Week Data Science Course

RESEARCH EXPERIENCE

Research Assistant for Mathematics and Computer Science Capstone *January 2023 – May 2023*
Integrative Ecology Lab, Dr. Jason Gleditsch, Dr. Matthew Helmus, Center for Biodiversity, Temple University *Philadelphia, PA*
- Analyzed species richness on 835 islands in the Greater Caribbean, exploring 54 biogeographic and anthropogenic variables at various spatial and taxonomic scales to understand the ecological mechanisms that produce them, their effect on an island's trajectory, and their influence on natural relationships
- Built and compared 30+ machine learning models to achieve more realistic predictions of species richness than traditional statistical processes and to identify the most influential drivers of biodiversity in the Anthropocene. Found ensemble-based learning methods to yield the highest performance with accuracy scores reaching up to 76.85%
- Created a relational database using Python and SQLite to easily extract, query, and aggregate 11 datasets for our biogeographic model

iEcoLab Research Assistant *January 2022 – May 2022*
Integrative Ecology Lab, Dr. Sebastiano De Bona, Dr. Matthew Helmus, Center for Biodiversity, Temple University *Philadelphia, PA*
- Developed a robust cloud infrastructure capable of aggregating and storing 350,000 datapoints of Spotted Lanternfly data, enabling seamless deployment of the resulting harmonized dataset for federal and state collaborators (USDA and PDA) and facilitating modeling of future spread
- Leveraged advanced techniques in container deployment and continuous integration, successfully creating an efficient Docker container to execute data wrangling processes and to address challenges related to data storage

ICER Advanced Computational Research Experience for Students, Research Intern *May 2021 – August 2021*
Krishnan Laboratory for Computational Genomics and Biomedical Data Science, Michigan State University *East Lansing, MI*

- Wrangled heterogeneous datasets that capture ontological relationships between 30,000+ diseases and 20,000+ genes to discover novel disease-gene interactions through use of a NetworkX digraph and multiple Deep Learning models
- Visualized various performance metrics from our disease-gene prediction model to highlight improvement of our novel framework over a baseline
- Presented project discoveries at MID-SURE 2021 Symposium to 300+ students, faculty, and external audiences

Agarwal A, Yannakopoulos A, Krishnan A. "Exploration and Analysis of Gene-Disease Associations." Poster presented at the Mid-Michigan Symposium Undergraduate Research Experiences. Michigan State University. MI, USA. July 28, 2021.

CNN Research Assistant *January 2021 – May 2021*
College of Science and Technology, Temple University *Philadelphia, PA*

- Optimized local greeting card firm's website search tool by collaboratively constructing a Deep Neural Network to enhance customer accessibility and drive sales growth
- Learned how to use and design Convolutional Neural Networks through an online course at Stanford University in preparation for exploitation of deep learning techniques

Cloud/Edge Computing Research Assistant *August 2020 – December 2020*
College of Science and Technology, Temple University *Philadelphia, PA*

- Assisted graduate students in the development of an online job scheduler for the University to substantially reduce the average job finish time
- Researched into cloud/edge computing clusters and various machine learning techniques used in Dr. Jie Wu's laboratory

WORK EXPERIENCE

Vanguard – Machine Learning Engineer *November 2024 – June 2025*
Technology Leadership Program, Center for Analytics and Insights – AWM, The Vanguard Group *Malvern, PA*

- Built unified LLM evaluation framework for enterprise chatbots—developed DeepEval/RAGAS/Arize pipelines, custom G-Eval/DAG judges, and custom metrics—such as Data Explorer for structured scoring of chain-of-thought outputs
- Led enterprise adoption efforts: co-authored evaluation playbook, facilitated cross-functional working sessions, mentored peers on SageMaker and Arize, and delivered capstone demo that aligned stakeholders on best practices
- Elevated Gen-AI model observability across team pipelines (FRS, VoC, Wealth/Asset, Adhoc)—shortened issue detection cycles by implementing tracing, code-based and LLM-as-a-judge evaluators, customized metrics, monitors, and alerts
- Strengthened ML pipeline reliability—standardized CI/CD orchestration, improved testing and rollback safeguards, and tightened experiment tracking to reduce operational risk and accelerate releases

Vanguard – Data Scientist and Quantum Computing Researcher *November 2024 – June 2025*
Technology Leadership Program, Center for Analytics and Insights – PI Engineering, The Vanguard Group *Malvern, PA*

- Led Workflow Automation project, streamlining workflow visualization and enhancing project planning for developers. Successfully delivered Gen-AI powered UI, chatbot, workflow diagram, API Gateway, Model Governance FAQ, AlphaBot information, Lambda, SM Endpoint, and S3 storage
- Contributed to internal paper on Portfolio Optimization via Quantum Computing, organized Quantum hackathon, assisted with codebase, and conducted experiments with our model on simulation data
- Engineered the Friction Reduction System (FRS) pipeline, reducing client friction using agentic systems

Vanguard – AI/ML Experimenter *April 2024 – November 2024*
Technology Leadership Program, Personal Investor Technology – Conversational Channels, The Vanguard Group *Malvern, PA*

- Developed and deployed over 50 dialog tasks for multiple Kore AI chatbots for Vanguard's Retirement Journey, improving call-time efficiency by saving phone assistants ~9 seconds/call across thousands of interactions and reducing Average Hold Time for clients by 4.5+ minutes
- Facilitated agent readiness by integrating task-specific information and prompting clients to prepare needed materials
- Supported RDM Containment Infrastructure by providing documentation, integrating DC2 and RDM APIs with our lambda, writing unit tests, updating the RMD status chatbot, and supporting team access to new technology requests like Cyara

Vanguard - Data Engineer *August 2023 – April 2024*
Technology Leadership Program, Chief Data Analytics Office – Corporate Services, The Vanguard Group *Malvern, PA*

- Supported development of comprehensive data lake to cater to diverse data and analytics needs of key business units. The data lake facilitates cross-enterprise, self-service insights, offering a holistic view of Vanguard's risk posture
- Led the Archer Content API (CAPI) Initiative, a modernized way of ingesting data from Ballast Point, replacing the outdated and inefficient Enhanced Reports. This initiative decreased time to insights, increased available data and data refreshes, and saved ~$6000 in costs, empowering risk teams to proactively present data through a risk lens to business leaders
- Elevated codebase quality by implementing test coverage analysis (for entire pipeline), unit tests, and security enhancements
- Successfully ingested 10+ data blocks and subforms using Python, AWS, SQL, and other services

- Coordinated with business partners/stakeholders to query new data and to address requests for new data functionality

Vanguard - Application Developer *June 2022 – August 2023*
College-to-Corporate Program, Global Investment Financial Systems, The Vanguard Group *Malvern, PA*

- Developed over 15 Access Monitoring Tableau visuals and dashboards for nearly 180 usage datasets to highlight the untapped potential of this data to division leaders and to illustrate their capability to save the company millions of dollars
- Leveraged Tableau API and server to automate data refreshing & publishing of workbooks, ensuring timely monitoring of user access for auditing purposes
- Reverse engineered 11 Excel Power Queries—representing weekly reports from Service Now—and created Python scripts to clean datasets of over 45,000 data instances using APIs and lambda functions with CI/CD tools, removing single point of failure and streamlining data cleaning process
- Collaboratively developed front-end of stock market simulation tool—using ReactJS and real-time stock data from Vanguard APIs—that educates Vanguard crew on safe and controlled investing and advances their stock trading skills

Teaching Assistant *January 2021 – May 2021*
College of Science and Technology, CIS 1166: Mathematical Concepts in Computing I *Philadelphia, PA*

- Supported the Professor in a Discrete Mathematics course through the daily management of over 27 students
- Graded quizzes and exams, led and facilitated discussion in Recitation sections, and held office hours weekly to ensure effective student learning and academic progress

Technical Support Head *June 2018 – June 2019*
YMCA of Greater Brandywine *West Chester, PA*

- Voluntarily founded and led a group of 24+ older adults by holding weekly classes in modern technology, demonstrating new iOS features and MacOS technologies to enhance their digital literacy
- Created support documentation that empowered and enabled community members to independently expand their skills and troubleshoot issues, reducing reliance on the support team for resolutions
- Initiated problem-resolution program to ensure optimal customer satisfaction after class hours, enhancing the learning experience for community members

SELECTED PROJECTS

Flight Delay Prediction at Scale | Git: github.com/aagarwal17/datasci_261_ML_At_Scale | *September 2025 – December 2025*

- Led end-to-end development of flight delay prediction system processing 31.1 million flights (2015-2019) using PySpark, implementing custom joins, cleaning raw data, and engineering 112 features (network centrality, interaction terms, etc)
- Conducted comprehensive EDA revealing temporal, carrier, and geographic delay patterns, scaling pipeline 6x while reducing missing data from 49.39% to <1% through checkpoint management and 3-tier weather imputation strategy
- Trained multilayer perceptron (MLP), Random Forest, and Gradient-Boosted Trees with temporal validation splits, achieving F_2-score of 0.73 with 82.87% recall through 40:60 undersampling strategy optimized for operational delay detection

Building Energy Consumption | Git: github.com/mids-w203/datasci-203-final-project | *November 2024 – December 2024*

- Developed multiple regression models in R to analyze Site Energy Usage Intensity (Site EUI) across diverse building types, incorporating log transformations and robust standard errors to address skewness and heteroskedasticity
- Applied statistical techniques including Pearson's correlation analysis, variance inflation factor (VIF) calculations, and Breusch-Pagan tests to assess predictor significance and validate model assumptions
- Utilized geospatial data exploration to analyze the impact of elevation, extreme temperatures, and building size on energy consumption, identifying key predictors through exploratory data analysis and visualization
- Generated insights for sustainable building design, confirming that higher elevation and extreme heat increase energy consumption, while colder temperatures reduce it, guiding decisions on insulation, HVAC systems, and energy strategies

AGM Food Delivery & Pick-Up Expansion | Git: github.com/mids-w205/project-3-aagarwal17 | *July 2024 – August 2024*

- Developed a data-driven expansion strategy using SQL, Python, and NoSQL (MongoDB, Redis) to analyze customer density, optimize pick-up locations, and improve delivery efficiency via the BART public transportation system
- Applied graph theory algorithms in Neo4j (degree, closeness, and betweenness centrality) to identify optimal BART stations for pickup locations, reducing delivery times and lowering operational costs
- Designed a hybrid delivery model integrating public transit-based pickup, drones, and autonomous robots, enhancing accessibility while reducing peak-hour delivery fees and environmental impact
- Improved customer reach and brand awareness by strategically selecting pick-up stations, leading to increased sales, lower logistics costs, and a scalable framework for future expansion

Artificially Creative | Git: github.com/airicbear/cis-4496-project | Website: cis-4496-project.vercel.app | *January 2023 – May 2023*

- Collaborated with two team members in Kaggle's "I'm Something of a Painter Myself" Competition, which involved translating 7028 photographs into Monet-style paintings using Generative Adversarial Networks (GANs), for our Capstone

- Developed web interface that allows users to perform image-to-artist translation in the style of Van Gogh, Cezzane, and Ukiyo-e. Also implemented artist-to-artist style transfer by leveraging the bidirectional nature of CycleGANs
- Placed in top 10 of all teams competing (MiFID of 39.7320) by using a CycleGAN with 30 epochs, Adam optimizer with Loss rate of 0.00002 and Beta 1 of 0.5, batch size of 4, and implementation of data augmentation and label smoothing
- Visualized RGB distribution and brightness of photos, paintings, and translated images to verify style and color transfer
- Coordinated with other team members by contributing to front-end design, debugging code, dividing work effectively, and scheduling and leading weekly Scrum meetings, ensuring successful completion of all deliverables within client deadlines

NLP Disaster Tweets | Git: github.com/aagarwal17/NLPDisasterTweets/tree/main | *January 2023 – May 2023*

- Participated in Kaggle's Disaster Tweets Competition, which involved classifying 10,000 Twitter tweets as fake or real natural disasters using various NLP techniques as well as implementing deep learning models for text-based data
- Ranked in the top 25% among 1,228 participating teams using a BERT Model with 12 hidden layers, hidden size of 768, and 12 attention heads; also used 2-Fold StratifiedKFold and FullTokenizer for tokenization to achieve F1 Score of 0.812
- Implemented various textual preprocessing techniques, including back translation, username expansion, sample relabeling, abbreviation/slang expansion, and complex/irrelevant character removal such as emojis, punctuation, and URLS
- Prepared project reports and consistently documented code and weekly improvements to facilitate effective presentation of new deliverables to the Professor during Scrum meetings

BettR Sports Betting | Git: github.com/cis3296f22/bettR | Website (NA): bettrsportsbetting.com | *August 2022 – December 2022*

- Designed an open-source information aggregation platform to visualize NBA Game Statistics with Shiny to provide calculations for breakeven spread and expected gain/loss on user-inputted bet as well as supply statistically informed insights into user sports betting history
- Built interactive dashboards that display model performance over time from multiple web sources for comparison and integrated educational resources that can explain sports betting and statistics to novices
- Developed application backend by engineering web-scrapers in R and Python to extract, transform, clean, and merge NBA game statistics from ESPN and FiveThirtyEight, which facilitated the implementation of data into the front-end
- Implemented automated testing for backend scripts, ensuring reliability during data wrangling
- Coordinated with other team members by contributing to front-end development, debugging code, dividing work effectively, leading weekly Scrum meetings, and maintaining project documentation, resulting in successful meeting of client deadlines

Owl Hacks Hackathon – Philadelphia Bail Fund | Git: github.com/aagarwal17/Owl-Hacks-Hackathon | *October 2020*

- Collaborated with two teammates over a two-day virtual hackathon to clean, visualize, and analyze almost 17,000 data instances for the Philadelphia Bail Fund to identify valuable patterns and gain insights
- Showcased the generated visuals and explained findings to judges and fellow competitors, which were evaluated based on criteria such as value added and readability

Brick Breaker | Git: github.com/aagarwal17/Brick-Breaker-Game | *January 2019 – June 2019*

- Developed a fully functional Brick Breaker Game in Java as the final project for my AP Computer Science course, serving as the culmination of everything I learned in my first year of programming
- Incorporated various modifications to the traditional Brick Breaker such as different game modes and power-ups, demonstrating my capability to apply learned concepts to create a dynamic gaming experience
- Designed the game with captivating graphics using Java's AWT and Swing libraries to enhance the visual interface of the game

LEADERSHIP

Intern Coach, Vanguard – Malvern, PA *June 2025 – August 2025*

- Served as summer intern leader—built a repeatable onboarding and coaching cadence, provided hands-on guidance and clear expectations, and fostered inclusion and momentum—facilitating smoother delivery of interns' responsibilities

TLP Finance Committee, Vanguard – Malvern, PA *January 2024 – August 2025*

- Actively contributing to the Technology Leadership Program Finance Committee as a board member, participating in board meetings and playing a key role in the creation and enhancement of our lectures and courses on personal finance

TLP Digest, Vanguard – Malvern, PA *January 2024 – August 2025*

- Actively contributing to the Technology Leadership Program Digest Committee as a board member, playing a key role in the creation and enhancement of our newsletters by headlining interviews with senior leadership and overseeing data maintenance

Intern Hackathon Technical Co-Lead, Vanguard – Malvern, PA *June 2024 – August 2024*

- Co-led Intern Hackathon, preparing extensively, conducting project independently, facilitating scrums, encouraging healthy debates, delegating tasks, facilitating Scrum calls, and providing feedback, emphasizing leadership and mentoring skills

Treasurer of Association for Computing Machinery (ACM), Temple University – Philadelphia, PA *September 2019 – May 2023*

- Updated membership list, collected membership fees, managed the expense report and financial records, kept track of all payments/receipts, managed the bank account, oversaw organization budget (for events, food, merchandise, etc.), and completed necessary forms to secure allocations, grants, and funding

IT Director/STARS Coordinator of American Statistical Association, Fox School of Business – PA *October 2019 – May 2023*
- Completed STARS requirements to allow the student professional organization to run effectively; updated and enhanced all website sections and social media pages; and promoted responsibility, respect, and positive experiences for all members

President of Habitat for Humanity, Temple University – Philadelphia, PA *October 2019 – May 2022*
- Supervised the Executive Board's planning of semester fundraising, build, and volunteer events; scheduled general body and Executive Board meetings; raised $200+ across 3 fundraisers; and fulfilled most executive roles including Secretary, Treasurer, STARS Coordinator, and Temple Student Government Leader due to a shortage of officers during the COVID-19 pandemic

Owl Hacks Hackathon Logistics Department Co-Leader, Temple University – Philadelphia, PA *August 2021 – October 2021*
- Voluntarily led the logistics department for the Owl Hacks 2021-2022 virtual coding competition

Webmaster of Temple University Data Science Community (TUDSC) – Philadelphia, PA *January 2021 – August 2021*
- Developed the website for a new organization that supports the academic and professional advancement of students in the Data Science field as well as related STEM areas

Sustainability Representative of 1300 Residence Hall, Temple University – Philadelphia, PA *September 2019 – May 2020*
- Served as the official voice for the residential student population; organized, managed, and led nearly a dozen events in a two-semester span in collaboration with a community council team to bring about a better community; and worked on building and demonstrating sustainability initiatives to promote a greener campus

COMMUNITY ENGAGEMENT

VanGarden, Vanguard – Malvern, PA *June 2024 – Present*
- Supporting company's garden, cultivating various fruits and vegetables that are donated to local organizations

TLP and C2C Internship Expo, Vanguard – Malvern, PA *June 2024*
- Co-Led Expo, providing advice to interns and answering questions on Data Science, master's programs, and TLP Program

WILS High School Career Exploration Program, Vanguard – Malvern, PA *August 2024*
- Welcomed 52 High School Interns on Vanguard campus to hear from various Vanguard leaders; assisted program setup

Community Service Association, Temple University – Philadelphia, PA *September 2019 – May 2020*
- Participated in various service activities, such as serving various dishes to 100+ homeless individuals at the Grace Café

Temple Community Garden Volunteer, Temple University – Philadelphia, PA *August 2019 – May 2020*
- Cultivated and nurtured various vegetables in communal beds, fostering an environment for innovation and experimentation within urban agriculture

Honors Connect Volunteer – Philadelphia, PA *August 11-18, 2019*
- Engaged in service projects with different community-based organizations such as MANNA to become acquainted with Temple University and the Philadelphia area

Volleyball Team Captain, Temple University – Philadelphia, PA *August 2019 – December 2020*
- Recruited members, managed dues and group contact, organized practice sessions, and led a dozen undergraduate students through multiple Intramural Volleyball games

Sustainability Task Force, Temple University – Philadelphia, PA *August 2019 – May 2020*
- Fostered a green campus, encouraged useful and forward-looking change, promoted environmental literacy, and made recommendations to the President to play a leading role in the environmental sector of my local community

Green Council, Temple University – Philadelphia, PA *August 2019 – May 2020*
- Represented student organizations by sharing ideas, planning and promoting events, and collaborating on a strategic vision for a more sustainable Temple and Philadelphia community

AFFILIATIONS

Science Scholars Student – Research program for STEM students preparing for graduate school *August 2020 – May 2023*
Undergraduate Research Student – Program allowing students to work with distinguished faculty *January 2021 – May 2023*
Math Club - Scholarly community of students and faculty with a passion for mathematics *August 2019 – May 2023*
Temple University Internship Student – Selective Program offering on-campus internship opportunities *July 2021 – May 2022*
DOW Diamond Symposium Attendee – Virtual interactive experience to engage with DOW leaders *August 16-18, 2021*

AWARDS/HONORS

Dean's List – Academic recognition given to the top 16% of all College of Science and Technology students *Fall 2019 – May 2023*

Distinction in Double Major – Academic recognition given for completion of special requirements for both majors *May 2023*
Summa Cum Laude – Top 2% of all baccalaureate recipients from Temple School of Science and Technology *May 2023*
Phi Beta Kappa Society – Inducted into oldest undergraduate honor society for the liberal arts and sciences in the U.S *May 2022*
President Student Service Award – 100+ hours of volunteer service in one year *May 2019*
Temple University Presidential Scholarship – Full tuition merit award *November 2018*